CSci 14 Lab Practical Exam

Largest Exponential Digit Sum

The number 10^{10} is a large number, however the sum of its digit is just 1.

Given natural numbers

$$a^b, where\ 0 \leq a \leq 9,$$
$$0 \leq b \leq 18$$

Implement the function below:

- `long long int intExp(int a, int b) -` returns the integer result of `a^b` using an iterative structure .
- `int digitSum (long long int x)` – returns the digit sum of a `long long int x`
- `void storeLargestExpDigitSum (int x[], int s)` – stores the largest sum of value for `a^b` into every element of array x using `long long int intExp(int a, int b)` and `int digitSum (long long int x)` functions. Example: `x[0]`: {max digit sum of `0^0, 0^1, 0^2....0^18`}, `x[1]`: {max digit sum of `1^0, 1^1, 1^2....1^18`}. Limit the size of `s` to 10.
- `int dispExpDigitSum(int x[],int s)` – content of `int x[]` with size of `s`